No reading avoids the height inequality: 36 readings of Corollary 3.12 and the conservation of the height load

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Abstract

Corollary 3.12 of inter-universal Teichmüller theory (IUT) III compares the log-volume $-|\log \underline{\underline{q}}|$ of the q-pilot object with the log-volume $-|\log \underline{\underline{\Theta}}|$ of the holomorphic hull of the union of the possible images of the Θ-pilot object. Theorem 1.10 of IUT IV bounds $-|\log \underline{\underline{\Theta}}|$ from above, and the two together give the height inequality. The value of $-|\log \underline{\underline{\Theta}}|$ depends on how the statement is read, along five axes: the range of the label permutations of (Ind1), container or iteration for (Ind3), the group of (Ind2), whether the hull is taken, and the archimedean container. Combining the axes found so far gives 36 readings that change the value.

We show that in each of the 36 readings, also with a bounded Kummer distortion added, Corollary 3.12 and the upper bound of IV together give the same height inequality $(k-1)|\log \underline{\underline{q}}| \le C$. The reason is that $|\log \underline{\underline{q}}|$, k and C are computed by one procedure that takes no reading; only the middle term $-|\log \underline{\underline{\Theta}}|$ depends on the reading.

Moreover, a reading decides only how much of the wall $(k-1)|\log \underline{\underline{q}}|$ is carried by Corollary 3.12 and how much by IV. This split is decided by the (Ind1)-axis alone; the other axes change only a bounded quantity that does not involve the q-orders. No reading makes both loads nonpositive. In the family obtained by replacing q with q^N, the statement that holds for every N carries no positive load, and the other carries at least the whole wall.

We prove these statements in Lean 4 for a component model (96 declarations, standard axioms only), and compute the 36 values for seven explicit initial Θ-data. No claim is made about the truth of Corollary 3.12, Theorem 1.10 of IUT IV, or the abc conjecture.

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1 Introduction

1.1 Motivation

The ultimate aim of this work is to understand exactly the algorithm by which IUT computes the height inequality. Earlier notes determined the value of one step of the computation of IUT IV that had been left open, the (Ind2)-term [Gap], and put the computation from an initial Θ-datum to the two sides of the height inequality into one program [WT].

That computation contains several choices that the text of the original papers does not fix uniquely; we call them *readings*. This raises a question: by choosing the readings differently, is there a route that avoids the height inequality, or one that makes the proof cheaper? The answer of this note is that no reading provides either.

1.2 The two statements and the readings

Corollary 3.12 compares two log-volumes. One of them is subject to the indeterminacies; the other is not.

> S "Write $-|\log(\underline{\underline{\Theta}})| \in \mathbb{R} \bigcup \{+\infty\}$ for the procession-normalized mono-analytic log-volume [i.e., where the average is taken over $j \in \mathbb{F}_l^*$ — cf. Remark 3.1.1, (ii), (iii), (iv); Proposition 3.9, (i), (ii); Theorem 3.11, (i), (a)] of the holomorphic hull [cf. Remark 3.9.5, (i)] of the union of the possible images of a Θ-pilot object [cf. Definition 3.8, (i)], relative to the relevant Kummer isomorphisms [cf. Theorem 3.11, (ii)], in the multiradial representation of Theorem 3.11, (i), which we regard as subject to the indeterminacies (Ind1), (Ind2), (Ind3) described in Theorem 3.11, (i), (ii)."
>
> "Write $-|\log(\underline{q})| \in \mathbb{R}$ for the procession-normalized mono-analytic log-volume of the image of a q-pilot object [cf. Definition 3.8, (i)], relative to the relevant Kummer isomorphisms [cf. Theorem 3.11, (ii)], in the multiradial representation of Theorem 3.11, (i), which we do not regard as subject to the indeterminacies (Ind1), (Ind2), (Ind3) described in Theorem 3.11, (i), (ii)."
>
> [IUT3], Corollary 3.12

The proof of Theorem 1.10 of IV bounds $-|\log \underline{\underline{\Theta}}|$ from above. In doing so it applies Proposition 1.4 (iii) of IV with the position of the theta value fixed at the label j:

S "We would like to apply Proposition 1.4, (iii), to the present situation, by taking" ... "'$i^{\dagger}$' to be $j \in \mathbb{S}^{\pm}_{j+1}$;" ... "'$\lambda$' to be 'ord$(-)$' of the element $\underline{\underline{q}}^{j^2}_{\underline{\underline{v}}_j}$" ... "if $\underline{v}_j \in \underline{\mathbb{V}}^{\mathrm{bad}}$"

[IUT4], proof of Theorem 1.10, Step (v)

This gives the chain

$$-|\log \underline{q}| \;\leq\; -|\log \underline{\underline{\Theta}}| \;\leq\; U(D),$$

whose two ends together give the height inequality. How the middle term $-|\log \underline{\underline{\Theta}}|$ is formed depends on the range of the possible images, the container, the group, the hull and the archimedean part; such a choice is what we call a *reading*. For each reading there are two statements, Corollary 3.12 and the upper bound of IV. We show that no such pair avoids the height inequality, and that a reading only moves the content of the wall between the two statements.

1.3 Results

- Theorem 3.2 (conservation): in every reading, Corollary 3.12 and the upper bound of IV together give the same height inequality $(k-1)a \leq C$, where $a = |\log \underline{\underline{q}}|$. The sum of the slacks of the two statements equals the slack of the height inequality.
- Theorem 3.3 (the split of the load): with the split point x, Corollary 3.12 carries the load $(x-1)a$ and IV carries the load $(k-x)a$. Their sum is always the wall $(k-1)a$. The two loads are never both nonpositive, and one of them carries at least half of the wall.
- Theorem 3.4 (the split point is decided by (Ind1) alone): in the component model the theta part is $-ka$ for V, increases in the order V, L, L0, and is homogeneous of degree one in the q-orders. The theta part of V equals those of L and L0 exactly when the q-orders agree at the places of positive weight above each prime.
- Proposition 3.5 (the other axes change only the remainder): (Ind2), (Ind3), the hull, the archimedean part and a bounded Kummer distortion change the value only by a bounded quantity that does not involve the q-orders.
- Theorem 3.6 (the proof cannot be made cheaper): in the family obtained by replacing q with q^N, no reading keeps the two statements together for all large N. In a reading in which IV holds for every N, the load of IV is 0 and Corollary 3.12 carries the whole wall. In a reading in which Corollary 3.12 holds for every N, one has $x \leq 1$, and IV carries at least the whole wall. The theta part is the slope, in N, of the values of the family.
- Example 3.7 (two places): above one prime there are a bad place of weight $w \leq \frac{1}{2}$ and a good place (the situation of Belyi points). In reading L0 one has $x \leq 1$. In V, IV always holds and Corollary 3.12 fails; in L0 it is the other way round.
- Section 5 (explicit data): the 36 readings are computed for seven initial Θ-data. All seven have a prime above which the q-orders differ, and x_{L} and x_{L0} are strictly less than k. For D_{211}, D_{109} and D_{15} one has $x_{\mathrm{L0}} = \frac{7}{8} < 1$.

In one sentence: a reading decides only which of the two statements carries the content of the wall; the total does not depend on the reading.

1.4 Relation to earlier work

- P Theorem 14.3 (ii) of the author's educational note [II] shows, in a model that treats the (Ind1)-axis alone, that in every reading Corollary 3.12 and IV together imply the height bound (an implication). There, what (Ind2) and (Ind3) add to the hull is represented

by one number E. This note extends that statement to five axes and the Kummer distortion, turns the implication into an identity (the sum of the loads), and adds the theorems that the split point is decided by (Ind1) alone, with the condition for equality, and the uniformity theorem for the family q^N.

- [x] Earlier, the public repository Takkun-kohinata/IUT_LEAN [TK] (GitHub, 15 July 2026) proved in Lean, for one reading, that the containment of Step (v) of IV, with $\lambda = \mathrm{ord}(q^{j^2})$, together with Corollary 3.12 gives the height inequality $(\ell(\ell+1)/12 - 1)\, a \le L$, with $a = |\log \underline{q}|$ as here and $L \ge 0$ its ledger of IV, Propositions 1.1–1.4 (`thetaHull_forces_height`); that conversely this inequality, together with the condition $C_\Theta \ge -1$ of IV, Theorem 1.10, gives a model with that containment (`thetaHullTranscription_iff`); and that with $L = 0$ the containment cannot hold (`no_lossless_theta_hull`), all in `IutLean/Multiradial/ThetaHullDichotomy.lean`. In the terms of this note this is the case $x = 1$, in which Corollary 3.12 carries no load and IV carries the whole wall.

- [x] The repository lana-agents/iut [lana] ("lana" below), a public Lean formalization, distinguishes two readings of (Ind1), the literal reading and the reading of IV, and states that the literal reading does not suffice for the ABC argument. It also proves that taking the hull for each (Ind1)-image separately gives the value of the reading of IV (`rhsDataOn_singleton` in `Iut/Concrete/Ind1Invariance.lean`).

  [x] "Under the literal union over the labels, the holomorphic hull at a tuple `c` is governed by `min_l ord_p(q_{c(l)})` rather than by the average of `ord_p(q_v)` over the places that IUT IV, Step (v), obtains via Proposition 1.7. So the literal reading does not suffice for the Theorem 1.10 / ABC argument: the points produced by the Belyi reduction have good and bad places above the same prime, where `min_l ord_p(q_{c(l)}) = 0`."

  [lana], `Iut/Concrete/Literal.lean`, module docstring

- [x] In the same repository, `Iut/WallEquiv/` collects formulations of the wall—the word of the interim report of Project LANA [LANA, §0.2]—and proves equivalences between them in its models, and `NonDerivable.lean` shows, for one reading, that Step (xi) does not follow from the formalized content of Theorem 3.11 alone. lana itself states that these equivalences hold only in the models.

  [x] "**Only in the models** (identified-copies output law, zero distortion — e.g. unit-induced Kummer identifications, `Iut/LogLink`): $1 \Leftrightarrow 2 \Leftrightarrow 3 \Leftrightarrow 5 \Leftrightarrow 6 \Leftrightarrow 10$. A formulation different from (SS) needs an η-algorithm whose output law is not `LGP` acting on an (Ind3)-choice, or a transport with nonzero distortion (a Kummer identification that is not unit-induced)."

  "**The missing input is ABC-strength**: `classicalABC_of_forall_xi` (Step (xi) for all genuine data gives `ClassicalABC`, via `classicalABC_of_thm311`)."

  [lana], `Iut/WallEquiv/README.md`

- [x] lana's file `Iut/Abc/P24Table.lean` (added on 14 October 2026) checks by `decide` a table of the ingredients that a proof of the Szpiro inequality must use (from a research note `P2P9.md`, §2.4, which is not in the repository). A "reading" there is a way of deciding the open entries of the table, not a reading of Corollary 3.12. For proofs through the IUT package (structure C6), the ingredient VA is added to every minimal hitting set, in the baseline reading of the table.

  [x] "`VA`: the IUT coupling `V ≥ (s - 1) a`."

  "With `C6` (proofs through the IUT package), `VA` is added to each minimal hitting set."

  [lana], `Iut/Abc/P24Table.lean`

- X The shape of H below, $k\,a \le a + C$, is that of the statement in Mochizuki's report on formalization that the multiradial algorithm leaves the height unaffected, up to a mild error, by multiplication by N [Form, §3]. The Lean port of the toolkit motizukipdf of D. Richards [Richards] (October 2026) writes this as $N \cdot h \le h + C \Rightarrow h \le C/(N-1)$ (`height_bounded_of_dilation`) and proves, from the comparison in the common container taken as a hypothesis, the bound $|\log \underline{q}| \le H/(N-1)$ with $N = \ell(\ell+1)/12$ and H its estimate of the hull term (`corThreeTwelve_of_thmThreeElevenFive`).
- H Thus the fact that proofs through the IUT package use the wall itself is already in Takkun-kohinata/IUT_LEAN and in lana, for one reading (`ThetaHullDichotomy.lean`, `NonDerivable.lean` and VA above). What this note adds is the form over the whole family of readings. lana does not state that in every reading the load only moves between the two statements while its total stays the same. For the literal reading lana states that it does not suffice, but not that the load moves to IV (checked at commit `553076a`, the head of the repository on 5 November 2026).
- X The correspondence with the dilemma of Scholze and Stix [SS] (blurring too much gives an empty statement, blurring too little a false one) is discussed in Section 6.2.
- X Joshi holds that the Tate parameter changes with (Ind1), (Ind2) and (Ind3) [Joshi]. This denies the premise of this note that the left end takes no reading (Section 6.3).
- P The author's note [W311] uses the same family $q \to q^N$ in its models of the log-volume content of the results cited in the proof of Corollary 3.12 (its Theorems 14, 15 and 22).
- N What we consider new, to the author's knowledge: the identity of the loads over the 36 readings and the Kummer distortion; the fact that the split point is decided by (Ind1) alone, the other four axes changing only the remainder; the uniformity theorem for the family q^N (a statement that holds for every N carries no positive load); and the computation of the 36 readings on explicit data. The condition for equality is not new in itself: lana's `Literal.lean` states, from revision `2b2190d` (11 October 2026), that the regions of its two readings coincide when the q-orders agree at the places above each prime, and that two bad places above p with different q-orders already break this; and [II, Proposition 13.2] gives the same condition for V and L0 in a model of the (Ind1)-axis alone. What is added here is the same condition for L, and its validity in all 36 readings. That the wall is needed is not new (Takkun-kohinata/IUT_LEAN and lana, above).

1.5 What is not claimed

- It is not claimed that the 36 readings are all the readings of the original papers. They are the combinations of the axes found so far.
- That each axis changes only a bounded quantity (Proposition 3.5) rests on a paper argument from the formula for each axis and on checks by the programs. Lean uses it as a hypothesis.
- The family obtained by replacing q with q^N is a model, not a family of actual curves.
- We do not decide which reading the original papers intend. Nothing is claimed about the truth of Corollary 3.12, Theorem 1.10 of IUT IV, or the abc conjecture.

1.6 Labels

S a statement of the original papers of IUT or of Mochizuki's documents. X a statement of others (lana, Scholze–Stix and others). P a statement proved here; P alone marks an elementary fact. R known mathematics, applied here in a new way; the known mathematics is

named. N a result new in this note, to the author's knowledge. H a reading or interpretation, not a proof.

2 Setting

2.1 The two statements and the height inequality

For an initial Θ-datum D (with a prime $\ell \geq 5$ and $\ell^* = (\ell-1)/2$) put:

- $a(D) = |\log \underline{\underline{q}}| > 0$: the negative of the log-volume of the q-pilot object;
- $k = \dfrac{1}{\ell^*}\displaystyle\sum_{j=1}^{\ell^*} j^2 = \dfrac{\ell(\ell+1)}{12}$: the average of j^2 over the labels; $k > 1$ for $\ell \geq 5$;
- $U(D)$: the upper bound of $-|\log \underline{\underline{\Theta}}|$ given by Steps (v)–(vii) of the proof of Theorem 1.10 of IV. As in the original, the contributions of the primes 2, 3, 5 and ℓ are bounded by Step (ii) and (R4). It is computed by the program `iut_height.py` of this note, which implements Steps (iv)–(viii) as written;
- $C(D) = U(D) + k\,a(D)$: the upper bound of IV without the term $-k\,a$ of the theta values.

Definition 2.1 (The two statements and the height inequality)**.** Write $M_R(D)$ for the value of $-|\log \underline{\underline{\Theta}}|$ in a reading R, and put

$$\begin{aligned} \mathrm{Cor}_R(D):&\ \ -a(D) \leq M_R(D),\\ \mathrm{IV}_R(D):&\ \ M_R(D) \leq U(D) = -k\,a(D) + C(D),\\ \mathrm{H}(D):&\ \ (k-1)\,a(D) \leq C(D). \end{aligned}$$

$\mathrm{H}(D)$ contains no reading.

P The height inequality H is the inequality $H/6 \leq F_\chi U$ of Proposition 3 of the author's fifth report [Fifth], written in other variables: there H is the degree of the divisor of the q-parameters, so that $a = H/(2\ell)$, and $\chi = 12/(\ell(\ell+1)) = 1/k$, $F_\chi = 1/(1-\chi) = k/(k-1)$ and $C = \frac{\ell+1}{4}U$ (the U of that report is not the $U(D)$ of this note).

H We treat the upper bound $U(D)$ of IV as one number that does not depend on the reading. What the text of IV computes is this one number, and when the reading is changed, the question is whether that number bounds the hull in the reading R (the statement IV_R of Definition 2.1). Recomputing an upper bound for each reading is treated in Theorem 3.3 (v).

P The quantities a, k, U and C are computed by one procedure that takes no reading (a and the side of Corollary 3.12 by the program of [WT], U by `iut_height.py`). That the left end is not subject to the reading is the wording of Corollary 3.12 itself (Section 1.2), and the procedure for the right end fixes the position $i^\dagger = j$ in Step (v). Moreover, we compared the procedure with an elementary computation written from the statement of Theorem 1.10 alone, without the machinery of IUT (`elementary_height.py`, which does not use the program of [WT]; both programs are the author's). For all seven data the places counted and the four numbers at the ends ($\log \mathfrak{q}$, $\log \mathfrak{f}$, $\log \mathfrak{d}$ and the right-hand side of the theorem) agree (`compare_elementary_iut.py`).

2.2 Five axes and 36 readings

Axis	Choices	Source
(Ind1): the factors on which the theta value of label j may sit	V: the factor j only; L: $1, \dots, j$ (permutations fixing 0); L0: $0, \dots, j$ (all permutations)	S V is Step (v) of IV. X V and L0 are lana's reading of IV and its literal reading. L is (L) of the catalogue [Cat]. P L0 also moves the label 0: fixing $\mathcal{D}_0$ does not fix the label 0 inside the larger capsules [Fourth, §6]
(Ind3)	container: the whole tensor product $\bigotimes \mathcal{I}$ of log-shells as support; iteration: only units and the pure tensors of the actual images of the same number of iterated log-links	S The container is IV, Proposition 1.4 (iii) and Step (v). H The iteration is our reading
(Ind2)	units: $\mathbb{Z}_p^\times$ and the field automorphisms; lattice: all $\mathbb{Z}_p$-linear automorphisms preserving the log-shell of each factor	X The lattice group is lana's `tensorAut`. P The group of IV, Proposition 1.2 gives the same value [Gap, §6.2]. P At a finite extension of $\mathbb{Q}_p$ the group Ism of IUT III, Theorem 3.11 (i) is $\mathbb{Z}_p^\times$ (by [MT, Thm. F, Thm. 7.4]), so (Ind2) acts by unit scalars [Fourth, §6, Proposition 9]
Hull	yes: the union is replaced by its holomorphic hull; no: the union itself	S Corollary 3.12 takes the hull. X No hull is lana's `SSIneqNoHull` ("not IUT's container", in its docstring)
Archimedean part	small: $\pi^{j+1} \cdot B_I$; large: the unit ball of the tensor-product metric	S Small is Step (vii) of IV. X Large is what lana keeps, following IUT III, Proposition 3.2 (ii) (`VcI.md`, §2(e))

In the iteration reading without the hull, the possible images form a union of pure tensors, of volume 0 (log-volume $-\infty$). Leaving out these combinations, there are $3 \times (2 \times 2 + 2) \times 2 = 36$ readings (Lean: `card_admissible`).

> S "Then it follows from Proposition 1.5, (iii), (iv), that $\pi^{j+1} \cdot B_I$ serves as a container for the 'union of possible images of a Θ-pilot object' discussed in Step (iv)."
>
> [IUT4], proof of Theorem 1.10, Step (vii)

H In addition, the Kummer distortion δ (the difference of log-volumes carried by a Kummer identification that is not induced by units; the distortion in lana's `WallOn`) is treated separately, as a continuous parameter; δ is added to M_R. Readings that do not change the value but decide whether the comparison applies at all, such as ($\mathrm{M_b}$) and ($\mathrm{M_g}$) of the catalogue [Cat], are not counted as axes.

2.3 The component model

We set up the model in which the Lean proofs are carried out. Take a finite set V of places above one prime p, weights $w_v \geq 0$ with $\sum_v w_v = 1$, and q-orders $Q_v \geq 0$ (in units of $\log p$; zero except at bad places). For $j = 1, \dots, \ell^*$, a component for the label j is a tuple $t = (t_0, \dots, t_j) \in V^{j+1}$ of weight $w(t) = \prod_i w_{t_i}$. Write $s_i(t) = j^2 Q_{t_i}$ for the order of the theta value in the factor i.

Let the sets of factors allowed by the (Ind1)-axis be $P_{\mathrm{V}}(j) = \{j\}$, $P_{\mathrm{L}}(j) = \{1, \dots, j\}$ and $P_{\mathrm{L0}}(j) = \{0, \dots, j\}$ (Lean: `pos`). The hull of the images that put the theta value on these

factors is decided by the largest image. The theta part on a component is

$$h_r(t) = \max_{i \in P_r(j)} \bigl(-s_i(t)\bigr) = -\min_{i \in P_r(j)} s_i(t) \qquad (r \in \{\mathrm{V}, \mathrm{L}, \mathrm{L0}\})$$

(`thetaHull`), and its procession-normalized value is

$$\Theta_r = \frac{1}{\ell^*} \sum_{j=1}^{\ell^*} \sum_{t \in V^{j+1}} w(t)\, h_r(t)$$

(`thetaPart`). The degree of the q-pilot is $a = \sum_v w_v Q_v$ (`qdeg`). With several primes, the values for each prime are multiplied by $\log p$ and added (`thetaSum`, `qdegSum`).

R Known mathematics: [II, Setup 12.1 and Propositions 13.1–13.3], where the readings are given by sets of permutations. The hulls of V and L0 are those of S_{b} (the permutations fixing j) and S_{a} (all permutations).

3 Main results

Definition 3.1 (Theta part, remainder, split point, loads)**.** Write the value of a reading R as its theta part plus a remainder, $M_R = \Theta_{r(R)} + B_R$, where $r(R)$ is the (Ind1)-axis of R. The theta part is the part proportional to the q-orders; the remainder B_R collects the container, the unit parts, the rounding of the lattice, the defect of the hull, the archimedean part and a bounded Kummer distortion. With the split point $x_r = -\Theta_r/a$ put

$$\begin{aligned} \text{load of Corollary 3.12} &= (x_r - 1)\, a, & \text{load of IV} &= (k - x_r)\, a, \\ \text{room of Corollary 3.12} &= B_R, & \text{room of IV} &= C - B_R. \end{aligned}$$

H For one datum alone this split is only bookkeeping, since every M_R can be written as a theta part plus a remainder. What decides the split uniquely is the slope in the family obtained by replacing q with q^N (Theorem 3.6 (iv)).

Theorem 3.2 (Conservation)**.** P

Elementary. Lean: `Datum.conservation`, `Datum.slack_sum`

In every reading R, Cor_R and IV_R together imply H*. Moreover*

$$(M_R + a) + (U - M_R) = C - (k - 1)\, a,$$

so the sum of the slacks of the two statements equals the slack of the height inequality.

Theorem 3.3 (The split of the load)**.** P

Elementary. Lean: `Datum.cor_iff`, `Datum.iv_iff`, `Datum.load_sum`, `Datum.x_le_k`, `Datum.loadIV_nonneg`, `Datum.loadIV_eq_zero_iff`, `Datum.half_load_le_max`, `Datum.not_both_loads_nonpos`, `Datum.adapted_bound`

Assume that the theta part satisfies $\Theta_r \geq -k\,a$ (in the model this holds by Theorem 3.4 (ii)). Then:

(i) $\mathrm{Cor}_R \iff (x_r - 1)a \leq B_R$, *and* $\mathrm{IV}_R \iff (k - x_r)a \leq C - B_R$.

(ii) The two loads add up to $(k-1)a$, and the two rooms add up to C. H *is the sum of the two inequalities.*

(iii) $x_r \leq k$, and the load of IV is nonnegative. It vanishes only when $\Theta_r = -k\,a$.

(iv) The larger of the two loads is at least $(k-1)a/2$. In particular the two loads are never both nonpositive.

(v) *If the upper bound of IV is replaced by the bound* $M_R \le \Theta_r + \mathrm{hi}$ *that holds automatically in the reading* R *(where* $B_R \le \mathrm{hi}$*), the two statements together give only* $(x_r - 1)a \le \mathrm{hi}$*. Its coefficient* $x_r - 1$ *of* a *is smaller than the coefficient* $k - 1$ *of the wall by exactly the load of IV,* $k - x_r$*.*

Theorem 3.4 (The split point is decided by (Ind1) alone)**.** R

Known mathematics: [II, Propositions 13.1–13.3] (V and L0); for one label position per capsule, lana's `rhsDataOn_singleton`; the condition for equality of lana's two readings is stated in lana's `Literal.lean` (Section 1.4). Lean: `thetaPart_V`, `thetaPart_mono`, `thetaPart_ge`, `thetaPart_nonpos`, `thetaPart_smul`, `thetaPart_eq_V_of_const`, `thetaPart_V_lt_L0`, `thetaPart_V_lt_L`, `thetaSum_V`, `thetaSum_ge`, `thetaSum_smul`

In the model of Section 2.3:

(i) $\Theta_{\mathrm{V}} = -k\,a$*. That is,* $x_{\mathrm{V}} = k$*, and in reading* V *the load of IV is* 0*.*

(ii) $\Theta_{\mathrm{V}} \le \Theta_{\mathrm{L}} \le \Theta_{\mathrm{L0}} \le 0$*. That is,* $k = x_{\mathrm{V}} \ge x_{\mathrm{L}} \ge x_{\mathrm{L0}} \ge 0$*.*

(iii) *If the q-orders are multiplied by* $N \ge 0$*, then* Θ_r *and* a *are multiplied by* N*, and* x_r *does not change.*

(iv) *If the q-orders are equal at all places of positive weight, then* $\Theta_{\mathrm{L}} = \Theta_{\mathrm{L0}} = \Theta_{\mathrm{V}}$*.*

(v) *If two places of positive weight have different q-orders, then* $\Theta_{\mathrm{V}} < \Theta_{\mathrm{L0}}$*. If* $\ell^* \ge 2$*, then also* $\Theta_{\mathrm{V}} < \Theta_{\mathrm{L}}$ *(for* $j = 1$*,* L *is the same as* V*).*

For sums over several primes (with coefficients $\log p \ge 0$*), (i)–(iii) hold as well.*

Proposition 3.5 (The other axes change only the remainder)**.** P

Elementary, given the formula for each axis; the sources of the formulas are given below. In Lean a hypothesis (`Family.rest_lo`, `Family.rest_hi`).

In units of $\log p$*, let* ρ_u *be the outer radius of the log-shell,* ρ'_u *a lower bound for its inner radius,* $\mu_u = 1 + \delta_p - \rho_u$ *with* $\delta_p = [p = 2]$*,* $n = j + 1$*, and* $\kappa(X) = \max\{k : X \subseteq p^k \mathcal{I}\}$*. The values on a component are as follows.*

- *Container, units, hull:* $\sum_u \rho_u - \min_{i \in P} s_i$*.*
- *Container, lattice, hull:* $\sum_u \rho_u - \min_{i \in P} \kappa(\theta_i \mathcal{I})$*, where* $s_i - g \le \kappa(\theta_i \mathcal{I}) \le s_i$ *with a local constant* g *(*$g < 1$ *at small places).*
- *Iteration, units:* $-\big(\min_{i \in P} s_i + \min(0, \sum_u \mu_u)\big)$*.*
- *Iteration, lattice:* $\sum_u \rho_u - \min_{i \in P} \min(A_i, B_i)$*, where* $A_i = \kappa(\theta_i \mathcal{O}) + \sum_{u \ne i} \kappa(\mathcal{O}_u)$ *and* $B_i = n(1 + \delta_p) + \kappa(\theta_i \mathcal{I})$*.*
- *No hull (container): at least the value with the hull minus the shell defect, and at most the value with the hull; the shell defect is nonnegative and at most* $d_I + \sum_{I^*} (3 + \log e)/\log p$*.*
- *Archimedean part: the hull of the large ball exceeds the small container by* $\frac{j}{2} \log 2$ *for the label* j *(by* $\frac{\ell^* + 1}{4} \log 2$ *on average over the labels). Without the hull, the large ball is smaller by a further* $\log(N!)/(2N)$*, with* $N = 2^j$*.*

Each formula is $-\min_{i \in P} s_i$ *plus a quantity bounded above and below by local constants that do not involve the q-orders (*ρ*,* ρ'*, the constant* $a + b$ *of IV, Proposition 1.2,* μ*,* $n(1 + \delta_p)$ *and* $3 + \log e$*). Hence the theta part is decided by the (Ind1)-axis alone, the other axes change only the remainder* B_R*, and that change does not involve the q-orders. A bounded Kummer distortion also goes into the remainder. A distortion* $d \cdot a$ *proportional to* a *lowers the split point by* d *(and moves load to IV if* $d > 0$*).*

Sources of the formulas. X The value $\sum \rho_u \log p$ of the hull of the container is lana's `componentVol_leastHull_tensorLogShell_eq`. R The bracketing of κ for the lattice group is Lemma 4.1 and Theorem 3.3 of [Gap]. H The two formulas for the iteration reading are our computation in our reading. X The bound for the shell defect and the values of the archimedean part are lana's (`VcI.lean`, `VcI.md`, §2(e)). X Zeng [Zeng] gives in closed form the index of the integral structure $\bigotimes^n \mathcal{O}_K$ in the product of the rings of integers of the components of $\bigotimes^n K$, for a Galois extension $K/\mathbb{Q}_p$, the index that the hull introduces; it depends on the discriminant of $K/\mathbb{Q}_p$ and on n, not on the q-orders. Bounded here means: not involving the q-orders; the remainder does depend on ℓ and on the local fields.

Theorem 3.6 (The proof cannot be made cheaper)**.** N

Lean: `Family.no_uniform_reading`, `Family.iv_uniform`, `Family.cor_uniform`, `Family.tendsto_slope`, `modelFamily_scaled`

Let D^N be the datum with q replaced by q^N. Assume that a and the theta parts are multiplied by N, that $\Theta_r \geq -k\,a$ for every r (in the model this holds by Theorem 3.4 (ii)), that k and C do not change, and that the remainders lie in intervals $[\mathrm{lo}_R, \mathrm{hi}_R]$ that do not depend on N (Theorem 3.4 (iii) and Proposition 3.5). Then in every reading R:

(i) There is N_0 such that $\mathrm{Cor}_R(D^N)$ and $\mathrm{IV}_R(D^N)$ do not hold together for $N \geq N_0$.

(ii) If $\mathrm{IV}_R(D^N)$ holds for every $N \geq 1$, then $\Theta_r = -k\,a$ (the load of IV is 0), and

$$\mathrm{Cor}_R(D^N) \iff (k-1)\,Na \leq B_R(D^N)\,:$$

Corollary 3.12 carries the whole wall.

(iii) If $\mathrm{Cor}_R(D^N)$ holds for every $N \geq 1$, then $\Theta_r \geq -a$ (that is, $x_r \leq 1$), and $\mathrm{IV}_R(D^N)$ implies $(k-1)\,Na \leq C - B_R(D^N)$: IV carries at least the whole wall.

(iv) $M_R(D^N)/N \to \Theta_r$ as $N \to \infty$. The family decides how the value splits into theta part and remainder.

Example 3.7 (Two places: the situation of Belyi points)**.** P

Known mathematics: [II, Proposition 13.3 and Theorem 14.3 (iv), (v)]. Lean: `thetaPart_L0_two`, `twoPlace_x_L0_le_one`, `belyi_cor_L0`, `belyi_iv_L0_fails`, `belyi_iv_V`, `belyi_cor_V_fails`

Above one prime there are a bad place of weight w and q-order $Q_{\mathrm{b}} > 0$ and a good place of weight $1-w$. Then $x_{\mathrm{L0}} = A(w) = \frac{1}{\ell^*}\sum_{j=1}^{\ell^*} j^2 w^j$, and $x_{\mathrm{L0}} \leq 1$ if $w \leq \frac{1}{2}$.

Take the family with $w = \frac{1}{2}$, $Q_{\mathrm{b}} = 1$ and remainder 0. In reading V, if $C \geq 0$, IV holds for every N and Corollary 3.12 fails for every N. In reading L0, Corollary 3.12 holds for every N and IV fails for large N. The two readings put the load on opposite sides.

4 Proofs

Proof of Theorem 3.2. Adding $-a \leq M_R$ and $M_R \leq -k\,a + C$ gives $-a \leq -k\,a + C$, that is, $(k-1)a \leq C$. The identity is obtained by expanding. □

Proof of Theorem 3.3. Substituting $\Theta_r = -x_r a$ gives (i). (ii) is the sum of the two inequalities of (i). (iii) follows from $-k\,a \leq \Theta_r = -x_r a$ and $a > 0$. (iv) follows from the fact that the sum is $(k-1)a > 0$. For (v), Cor_R and $B_R \leq \mathrm{hi}$ give $(x_r - 1)a \leq B_R \leq \mathrm{hi}$; the relation between the coefficients follows from (ii). □

Proof of Theorem 3.4. (i) In V, $h_{\mathrm{V}}(t) = -j^2 Q_{t_j}$ depends on one factor only, so its weighted average is $-j^2 \sum_v w_v Q_v = -j^2 a$ (because $\sum w = 1$). Averaging over j gives $-k\,a$. (ii) Since $P_{\mathrm{V}} \subseteq P_{\mathrm{L}} \subseteq P_{\mathrm{L0}}$, the maximum on each component increases; since the weights are nonnegative,

so does the average. $Q \geq 0$ gives $h \leq 0$. (iii) Multiplication by a nonnegative number commutes with the maximum. (iv) On a component of positive weight all factors have the same order, so the maximum equals h_{V}; components of weight 0 do not enter the sum. (v) Take places u, v with $Q_u < Q_v$. The component (u, v) for $j = 1$ has weight $w_u w_v > 0$, and there $h_{\mathrm{L0}} \geq -Q_u > -Q_v = h_{\mathrm{V}}$; on the other components and labels one has $\geq$. For L use the component (u, u, v) for $j = 2$. □

Proof of Proposition 3.5. Each formula consists of a term of the form $\min_{i \in P}(s_i - c_i)$, with $0 \leq c_i \leq c_{\max}$ and $c_{\max}$ a local constant, and terms that do not involve the q-orders. Since

$$\min_i s_i - c_{\max} \leq \min_i (s_i - c_i) \leq \min_i s_i,$$

the remainder lies in an interval that does not depend on the q-orders. The readings without the hull and with the large ball add to or subtract from the value with the hull quantities that do not involve the q-orders. □

Proof of Theorem 3.6. (i) If $N > C/((k-1)a)$, then $\mathrm{H}(D^N)$ fails, so by Theorem 3.2 the two statements do not hold together. (ii) If $\Theta_r > -k\,a$, then $g = \Theta_r + k\,a > 0$, and $\mathrm{IV}_R(D^N)$ gives $Ng \leq C - B_R(D^N) \leq C - \mathrm{lo}_R$, which fails for large N. Hence $\Theta_r = -k\,a$, and substituting gives the equivalence. (iii) If $\Theta_r < -a$, then $g = -a - \Theta_r > 0$, and $\mathrm{Cor}_R(D^N)$ gives $Ng \leq B_R(D^N) \leq \mathrm{hi}_R$, which fails for large N. For the second half insert $\Theta_r \geq -a$ into $\mathrm{IV}_R(D^N)$. (iv) $M_R(D^N)/N = \Theta_r + B_R(D^N)/N$, and both ends of $\mathrm{lo}_R/N \leq B_R(D^N)/N \leq \mathrm{hi}_R/N$ tend to 0. □

Proof of Example 3.7. In L0 the maximum on a component is $-j^2 Q_{\mathrm{b}}$ if all factors are at the bad place and 0 otherwise. That event has weight w^{j+1}, so $\Theta_{\mathrm{L0}} = -wQ_{\mathrm{b}}A(w)$ and $a = wQ_{\mathrm{b}}$. A is increasing in w, and

$$\sum_{j=1}^{n} j^2/2^j = 6 - (n^2 + 4n + 6)/2^n \leq n$$

gives $A(\frac{1}{2}) \leq 1$. For V, Theorem 3.4 (i) and the remainder 0 give $\mathrm{IV} \iff 0 \leq C$ and $\mathrm{Cor} \iff (k-1)Na \leq 0$, which is false. IV in L0 follows from Theorem 3.6 (i) and Cor. □

5 Computations on explicit data

5.1 Procedure

P We use the seven initial Θ-data of the catalogue [Cat]. The computation uses the program of the walkthrough [WT], version 1.1.0, extended by the hulls of twelve readings (`iut_patterns.py`), their assembly into the 36 readings (`iut_readings36.py`) and the upper bound of IV (`iut_height.py`). As in [Gap] and [WT], the set $\mathbb{V}_{\mathrm{mod}}^{\mathrm{bad}}$ is that of the catalogue minus the places above 3 and 5, where the ramification of K depends on the construction of F (Definition 3.1 (b) of IUT I allows such a subset; see [Gap, §5]). The places above 2, 3, 5 and ℓ therefore have theta value 1, and so have those above the unfactored cofactor of D_{2357}. In the readings with the hull these places only add nonnegative amounts to M_R; in the readings without the hull they may also lower M_R, by at most their shell defect, which is not computed here. In the upper bound of IV they are bounded by Step (ii) and (R4), as in the computation of $U(D)$. In every reading the contribution of a prime at which the theta value is 1 is at most the bound of IV there (with theta value 1 the lattice group gives the same as the units, and the iteration and the readings without the hull are at most the hull of the container).

Table 1: The ends and the split points ($\ell = 7$ and $k = 14/3$ except for D_{2357}). The last column is the number of primes above which the q-orders differ at places of positive weight.

Datum	ℓ	a	wall $(k-1)a$	C	x_{V}	x_{L}	x_{L0}	primes
D_{211}	7	21.801	79.937	1247.14	4.6667	1.7500	0.8750	5
D_{109}	7	19.114	70.085	1187.52	4.6667	1.7500	0.8750	6
D_2	7	17.149	62.881	1235.19	4.6667	2.2273	1.4955	2
D_3	7	80.589	295.49	2247.61	4.6667	2.3149	1.6094	15
D_{18}	7	43.925	161.06	1683.07	4.6667	1.7779	0.9113	15
D_{15}	7	36.862	135.16	1556.68	4.6667	1.7500	0.8750	15
D_{2357}	157	11.840	24463.9	16229.6	2067.17	0.1538	0.0769	9

5.2 The split point

P For all seven data, $x_{\mathrm{V}} = k$ holds exactly as rational numbers ($\Theta_{\mathrm{V}} + k\,a = 0$ was checked with rational coefficients of the $\log p$). For D_{211}, D_{109} and D_{15}, above every bad prime there are exactly one bad and one good place, each of weight $\frac{1}{2}$. Hence $x_{\mathrm{L0}} = A(\frac{1}{2}) = \frac{7}{8}$ and $x_{\mathrm{L}} = 2A(\frac{1}{2}) = \frac{7}{4}$: this is exactly the situation of Example 3.7.

5.3 The verdicts in the 36 readings

P For the six data that are completely factored (all except D_{2357}):

- The upper bound of IV holds in all 36 readings.
- Corollary 3.12 holds for all six data in the container readings with the hull and in the readings of iteration with the lattice group (18 readings).
- In the readings of iteration with units (6 readings), Corollary 3.12 is short for all six data in V and L: the counted places fall short, and it fails unless the places that are not counted (those above 2, 3, 5 and ℓ) add at least the shortfall. In L0 only D_3 is short.
- In the container readings without the hull (12 readings) the verdict is undecided, because the shell defect is only bracketed. Only in L0 for D_{109} does it hold, for the counted places.

H That Corollary 3.12 and IV hold together on the actual data is because these data satisfy the height inequality with a large margin (for the six data the wall $(k-1)a$ is 5–13% of C). These verdicts test nothing. The effect of the reading becomes visible in the family q^N.

5.4 Replacing q by q^N

P The program multiplies only the q-orders by N (the ramification indices and $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ are kept, and $\ell \nmid N$). For the six data and $N = 2, 3, 25$ we checked that a and the theta parts are multiplied by N exactly, and that x and C do not change. For D_2, D_{211} and D_{109} we checked $N = 1, \ldots, 40$ (35 values, leaving out the multiples of ℓ). In every reading the range of the remainder B_R is at most 19.9, 25.9 and 22.7 respectively, about 3% of the growth of Na (669 to 850). The largest range is already reached at $N = 3$ and does not grow beyond it. For D_{109} the range 22.707 equals $\Delta(D_{109})$, the value of [Gap]: at $N = 3$ the numerators $n_u \in \{29, 1\}$ of the q-orders, quadratic residues modulo 7, become non-residues, so by [Gap, Proposition 3.6] the label average of the rounding term of the lattice group goes from $\frac{1}{3}$ to $\frac{2}{3}$ and the term doubles, from $\Delta(D_{109})$ to $2\Delta(D_{109})$ (in reading V with the container and the hull the remainder goes from 228.556 to 251.262).

A single number is the latest threshold over the twelve readings with the same (Ind1)-axis; a range runs from the least to the largest threshold over these readings (`thresholds.py`). In the readings of L0 without the hull the remainder has a negative lower end, so for D_{211}, D_{18}

Table 2: For which N each statement fails in the family q^N. The remainder B_R is that of the places counted in Section 5.1, replaced by its interval at $N = 1$.

Datum	H fails	V: Corollary 3.12	L0: Corollary 3.12	L0: IV fails
D_{211}	$N > 15.60$	fails for $N > 3.31$	holds for every N with the hull, and for $N \geq 2$ in all twelve	$N > 12.12$ to 15.15
D_{109}	$N > 16.94$	fails for $N > 3.27$	holds for every N	$N > 13.46$ to 16.34
D_2	$N > 19.64$	fails for $N > 4.29$	fails for $N > 29.83$	$N > 18.05$ to 22.95
D_3	$N > 7.61$	fails for $N > 3.14$	fails for $N > 17.67$	$N > 5.60$ to 9.30
D_{18}	$N > 10.45$	fails for $N > 3.41$	holds for every N with the hull, and for $N \geq 6$ in all twelve	$N > 7.11$ to 10.33
D_{15}	$N > 11.52$	fails for $N > 3.44$	holds for every N with the hull, and for $N \geq 3$ in all twelve	$N > 8.05$ to 11.22

and D_{15} Corollary 3.12 is guaranteed only from the stated N on. In V, IV carries no load and holds for every N. For each reading the smaller of its two thresholds is at most that of H (Theorem 3.2). Corollary 3.12 fails in V much earlier than H because its room B_R is much smaller than the whole room C of IV: 38–71% of C is the bound at the primes 2, 3, 5 and ℓ (Step (ii) for the different and (R4) for the terms of I^*), which is not in B_R. The places above these primes change M_R by an amount that does not depend on N: in the readings with the hull it is nonnegative and at most that bound, and without the hull it may also be negative, by at most their shell defect. Such an amount moves the thresholds of Corollary 3.12 and of IV in opposite directions and leaves that of H unchanged.

5.5 D_{2357}

P For D_{2357} ($\ell = 157$) the counted places alone leave the height inequality short by 8234 ($(k-1)a = 24464 > C = 16230$). In reading V this deficit appears on the side of Corollary 3.12 (IV holds); in L and L0 it appears on the side of IV (Corollary 3.12 holds or is undecided, and the deficit of IV is 8362 to 10308 in the readings with the hull and the small archimedean container, 8376 to 10321 with the large ball, and at least 7775 without the hull, where the shell defect is only bracketed). In every reading one of the two statements has a deficit.

This is conditional. The places above the unfactored cofactor of 1564 digits are not counted. In the readings with the hull they add nonnegative amounts to M_R, and they add to $U(D)$, where the estimate of Step (ii) at 2, 3, 5 and ℓ also grows with the conductor; so either deficit may be filled. Moreover, $U(D)$ is the upper bound of Steps (v)–(vii); compared with the final uniform form of IV (right-hand side 3.47×10^9), IV holds in every reading.

6 Discussion

6.1 Cannot be avoided, cannot be made cheaper

H The question had two parts: whether choosing the readings differently avoids the height inequality (avoidance), and whether it makes the proof cheaper (cost).

- *It cannot be avoided.* In every reading, the two statements together are the same height inequality (Theorem 3.2). This is because the ends take no reading, which rests on the wording of Corollary 3.12 and on Step (v) of IV (Section 2).
- *It cannot be made cheaper.* A reading only moves the content of the wall between the

two statements; the total $(k-1)a$ does not change (Theorem 3.3 (ii)). No reading makes both loads nonpositive (Theorem 3.3 (iv)). Making one statement hold for every N makes the other heavier by the same amount, so that it carries at least the whole wall (Theorem 3.6 (ii), (iii)). Rebuilding IV for each reading into a form that holds automatically weakens the resulting inequality by exactly that amount (Theorem 3.3 (v)).

H If IUT makes a contribution of its own to the height inequality, it should lie in computing the two sides by a method different from the elementary one. But in each of the 36 readings the two sides are computed by the same single procedure (Section 2), and the numbers at the ends agree with the elementary computation. What depends on the reading is only the middle term $-|\log \underline{\underline{\Theta}}|$, and it cancels when the two statements are combined.

6.2 Two typical readings

H In reading V the load of IV is 0, and Corollary 3.12 carries the whole wall: Corollary 3.12 is itself the height inequality $(k-1)a \le B_R$. This corresponds to the side of the dilemma of Scholze and Stix [SS, §2.2] on which the factor j^2 is not absorbed by any blurring: Corollary 3.12 is then a height inequality that has to be proved as it stands, and in the family q^N it fails for large N. The inequality $(k-1)Q \le E$ of Part I of [II] has this form.

H In reading L0, for data whose good and bad places lie above the same prime (Belyi points), one has $x \le 1$ and Corollary 3.12 carries no positive load: it holds whatever q is, and IV carries at least the whole wall. This is the situation of the option of Scholze and Stix labelled (c) in [II, Proposition 5.3 and §15.1], in which "the blurring must be by a factor of at least $\mathrm{O}(\ell^2)$ rendering the inequality thus obtained useless" [SS, §2.2]; this sentence has been formalized in Lean 4, with the constant $\ell(\ell+1)/12$, our k, in place of $\mathrm{O}(\ell^2)$ [sfingali]. The statement of lana that the literal reading does not suffice for ABC (Section 1.4) can be read as a description of this side from the standpoint of IV.

H The other readings put the split point x between these two. The point x is decided by (Ind1) alone, and for data whose q-orders agree, every reading has $x = k$, as V does.

6.3 What could change the conclusion

H The conclusion of this note changes only if one of the following happens.

- The ends depend on the reading. For the left end this would contradict the wording of Corollary 3.12 (the log-volume $-|\log \underline{\underline{q}}|$ is not subject to the indeterminacies). X It is the position of Joshi, for whom "the Tate parameter changes in non-trivial ways with changes of Arithmetic Holomorphic Structures and in particular with Ind_1, Ind_2 and Ind_3 (to be clear Mochizuki does not prove this–that is why the theory is incomplete)" [Joshi, (8) (iv)]; on that view the conservation of this note does not apply.
- The theta part is smaller than $-k\,a$. The possible images include the Θ-pilot object itself (the identity permutation and the identity automorphism), so the value of the hull is at least that of the Θ-pilot. A Kummer distortion proportional to a that acts in the negative direction, however, could give $x > k$. Even then the sum of the loads does not change (the proof of Theorem 3.3 (ii) does not use $\Theta \ge -k\,a$), and the load moves to the side of Corollary 3.12.
- A reading outside the 36 has a theta part that is not proportional to the q-orders. Then there is no slope (Theorem 3.6 (iv)), and the split of this note cannot be used.

7 What is not shown

- That the 36 readings are all the readings of the original papers.

- A Lean proof that each axis changes only the remainder (Proposition 3.5). Lean uses as a hypothesis that the remainders lie in intervals that do not depend on N, and defines the theta part as the maximum over the set of positions. That the axes other than (Ind1) do not change the theta part was checked by a paper argument from the formula for each axis and by the programs.
- That the family q^N is a family of actual curves.
- Which reading the original papers intend, and whether Corollary 3.12 holds.
- The verdict for D_{2357}, which depends on the unfactored cofactor.

8 Checks and reproducibility

The folder `reading-family/` of the repository `https://github.com/turahigashi/iut-explicit-initial-theta` (tag `v0.24.0`), also deposited with this note, contains the Lean files, the programs and their outputs.

- *Lean.* `lean/ReadingFamily.lean` (1024 lines) was checked with Lean 4.30.0 and Mathlib at tag `v4.30.0`. The file `lean/ReadingFamilyAxiomCheck.lean` is the same file followed by `#print axioms` for all 96 named declarations. Every declaration depends at most on `propext`, `Classical.choice` and `Quot.sound`; the definition of the readings, `Reading.Admissible`, depends on none. There is no `sorry`, no `axiom`, no `opaque` and no `native_decide`; there are no errors and no warnings. As a witness that the hypotheses are not empty, the two-place family `belyiFamily` is constructed.
- *Programs* (Python 3.10 or later with sympy; their comments and output are in Japanese, as for the program of [WT]). `scripts/iut_compute.py` is the program of [WT], version 1.1.0, unchanged. `iut_patterns.py` computes the twelve readings with the hull; with `--check` it confirms that its container–units readings agree with the original program, that its difference of the two (Ind2) models in reading V is $\Delta(D)$ of [Gap], that its closed formulas agree with the enumeration of the tuples, and that the intervals for κ contain the exact values in model fields. Its output is identical, character for character, whether it uses version 1.0.0 or version 1.1.0 of the program of [WT]. `iut_readings36.py` assembles the 36 readings, checks $\Theta_{\mathrm{V}} = -k\,a$ exactly, and checks numerically the order $\mathrm{V} \le \mathrm{L} \le \mathrm{L0} \le 0$ of the theta parts and the sums of the loads and of the rooms, for all data; with `--scale` it checks the family q^N. `scan_scale.py` scans $N = 1, \ldots, 40$, `thresholds.py` computes Table 2, `iut_height.py` computes $U(D)$, and `elementary_height.py` and `compare_elementary_iut.py` give the comparison with the elementary computation of Section 2.

Acknowledgements and use of AI

The question studied in this note—whether choosing the readings differently avoids the height inequality or makes the proof cheaper—arose in the author's discussions with generative AI tools; the author judged it important and decided to make it the subject of this note. Its ultimate aim, to understand exactly the algorithm by which IUT computes the height inequality, is the author's research direction. Generative AI tools (large language models) were used extensively, under the author's direction, in finding the proofs, in the computations, in the verification in Lean and in the writing. The author takes full responsibility for the contents of this note.

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